



REVIEW ARTICLE

Algal biostimulants as climate smart solutions for enhancing abiotic stress tolerance in crops

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Abstract

Climate change causes abiotic stresses like drought, salinity, temperature extremes and heavy metal contamination. These stresses threaten global agricultural productivity and food security. Chemical-based inputs may improve short-term yields, but they often harm soil health and increase environmental problems. In this situation, algal biostimulants from macroalgae, microalgae and cyanobacteria have emerged sustainable solutions for improving crop resilience to these stresses. These bioformulations are rich in polysaccharides, phytohormones, betaines, amino acids, phenolics, carotenoids and essential minerals. Together, they help plants respond better under stress. Algal biostimulants boost photosynthetic efficiency, strengthen antioxidant defences, regulate osmolyte accumulation and maintain ionic balance, which reduces oxidative damage and preserves cell integrity. They also enhance root structure, improve nutrient uptake and increase water use efficiency, supporting plant growth in extreme conditions. Recent evidence shows that they assist in priming stress-responsive genes, stabilise hormonal signalling and enhance stress memory for better adaptation. Furthermore, algal formulations improve soil health, boost microbial activity and reduce reliance on synthetic fertilisers, fitting into climate-resilient agricultural practices. Despite their several advantages, challenges remain in standardisation, dosage optimisation, understanding mechanisms and validating large-scale field applications. This review summarises existing knowledge on the mechanisms, formulations and specific applications of algal biostimulants. It emphasises their potential as eco-friendly solutions for improving tolerance to abiotic stress and fostering sustainable crop production in a changing climate.

Keywords: abiotic stress tolerance; algal biostimulants; microalgae and cyanobacteria; photosynthetic efficiency; rhizosphere interactions; seaweed extracts

Introduction

Agriculture in the 21st century is facing uncertainty in production due to accelerating climate change (1). As a result, global food security has become a significant concern. Temperature fluctuations, unpredictable monsoons, extended dry spells, erratic rainfall, salinity problems and recurrent heat waves have become more frequent (2). The normal functioning of plants is greatly affected by these stresses. These environmental fluctuations harm plant physiology by reducing photosynthesis, impairing enzymatic activity and causing nutrient imbalances and oxidative stress (3). These biochemical and cellular imbalances affect the metabolic efficiency and the reproductive development of plants, which eventually leads to a substantial reduction in yield. Drought, heat and other environmental stresses greatly reduce crop productivity. Drought can reduce the yield of wheat by about 21–40 % globally and maize by up to 40 % in case of severe drought. In severe cases, wheat and rice yield losses can be more than 50 % depending on the growth stage and stress intensity. Moreover, it is estimated that at the global scale, each 1 °C increase in temperature will reduce wheat yields by about 6 %. What these losses collectively mean is a serious threat to global food security (4).

The rise in population and the shrinkage of cultivable land emphasises the need to produce more food using the limited resources (5). Traditional agricultural approaches utilising synthetic fertilisers and pesticides cause nutrient leaching, soil organic carbon degradation, microbial imbalance and greenhouse gas emission (6). Precision agriculture and breeding climate-tolerant varieties are effective but have limitations of high cost and infrastructure requirements. In this regard, agriculture needs a new paradigm that uses inputs that are economically accessible, biologically efficient, safe for the environment and able to improve the inherent resilience of plants rather than only maintaining growth (7). Uncertainty in agriculture necessitates a transition from short-term yield maximisation to long-term resilience, sustainability and climate adaptability. This shift forms the foundation for climate-smart agriculture (CSA). The concept of CSA has emerged for improving crop productivity through adaptation to climate change while maintaining natural resources. The CSA promotes crop production by reducing dependence on chemicals and employing ecosystem-based management techniques (8, 9). Climate-smart agriculture is a set of practices such as integrated crop-livestock systems, agroforestry, conservation agriculture, improved nutrient and water

management and restoration of soil health to increase productivity, resilience and sustainability under climate change. Of these approaches, algal biostimulants are notable for their ability to promote plant growth, increase nutrient-use efficiency and enhance tolerance to abiotic stresses such as drought, salinity and temperature extremes, thus providing a multifunctional and efficient tool for climate-resilient crop production (9).

Biostimulants are a new class of plant-supporting inputs that can enhance growth, nitrogen (N) uptake and stress tolerance without acting as fertilisers. They activate stress-responsive pathways by altering the internal plant metabolism and triggering the hormone signalling pathways (10, 11). The nutrient utilisation efficiency is enhanced, root-shoot architecture is reinforced and antioxidant machinery is strengthened. This enables crops to withstand unfavourable conditions (12). Among the various types of biostimulants, those derived from microalgae and macroalgae (seaweeds) are emerging as effective and environmentally sustainable options, as evidenced by studies showing that foliar application of seaweed extract (0.5 %) significantly improves onion growth, yield and quality attributes, including increased bulb weight, total soluble solids (TSS) and essential nutrient content under field conditions (13). These algae produce a wide array of physiologically active substances, including phytohormones, phenolics, carotenoids, vitamins, minerals, peptides and antioxidants (14, 15). These metabolites help plants achieve homeostasis during stressful conditions by affecting multiple biochemical pathways at once. Research has shown that algal extracts can boost seed germination, enhance establishment and improve photosynthetic efficiency (16). By promoting the synthesis of osmolytes and controlling stomatal conductance under arid conditions, algal chemicals enhance water retention. Ionic equilibrium is preserved under saline stress by preventing sodium (Na^+) buildup and preserving membrane integrity (17). During temperature stress, they stabilise proteins and preserve hormonal balance by increasing the activity of antioxidant enzymes such as superoxide dismutase (SOD), catalase (CAT), peroxidase (POD) and ascorbate peroxidase (18). Algal biostimulants can help CSA by providing a safe, natural and biodegradable alternative to chemical input (19). The conceptual framework illustrating the role of algal biostimulants in CSA is presented in Fig. 1.

Despite their significant advantages, there remains a limited understanding of the mechanisms involved, field-level validation and standardisation of extraction techniques for algal biostimulants (20). This highlights the necessity to evaluate the existing information and the constraints that could accelerate the wider acceptance of algae-derived products. This review seeks to investigate the function of algal biostimulants as climate-smart inputs by enhancing crop resilience to stress, concentrating on their biological mechanisms, pathways for stress mitigation, practical applications and prospects for future research. This review presents an integrative and critical synthesis of algal biostimulants, highlighting mechanistic variability reported across studies and identifying inconsistencies in physiological and molecular responses under different crops, environmental conditions and formulation types. It also summarises emerging molecular insights while emphasising areas where current understanding remains limited. In addition, it identifies key research gaps related to dose-response relationships, genotype-specific responses and field-scale validation.

Literature search methodology

The literature used in this review was collected through searches conducted in Google Scholar and ScienceDirect. Publications relevant to algal biostimulants and CSA published between 2009 and 2026 were considered. The search included keywords such as "algal biostimulants", "seaweed extracts", "microalgae", "macroalgae", "CSA", "abiotic stress tolerance", "drought stress", "salinity stress", "soil health", "plant growth promotion", "algal biotechnology" and "sustainable agriculture". Original research articles, review papers, book chapters and relevant regulatory publications were selected based on their relevance to the objectives of the review. Additional references were identified from the bibliographies of selected articles. A total of 70 references were included in preparing this review.

Algal biostimulants

Types of algae used as biostimulants

Algae represent one of the most diverse biological groups used in sustainable crop enhancement. They can be broadly categorised into macroalgae, microalgae and cyanobacteria. Macroalgae, also known as seaweeds, are large marine algae that form the basis of

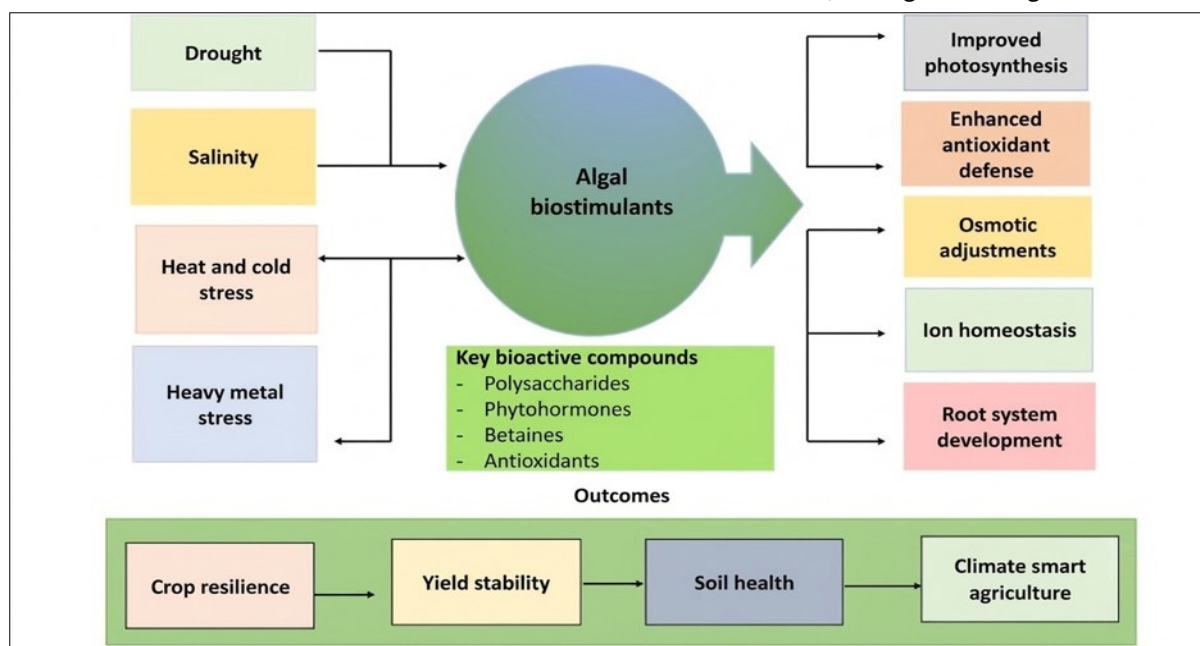


Fig. 1. Conceptual framework illustrating the role of algal biostimulants in climate-smart agriculture.

most commercially available biostimulant formulations (21). Liquid extracts and powders of species like *Ascophyllum nodosum*, *Sargassum* spp., *Ecklonia maxima* and *Laminaria digitata* have been widely used across cereals crops. Their biomass is naturally rich in phlorotannins, alginates, laminarin, mannitol, vitamins and essential minerals, which enhance nutrient absorption and plant vigour (22).

Macroalgae can be further divided into brown (Phaeophyceae), red (Rhodophyta) and green algae (Chlorophyta), based on their distinct metabolite reservoirs (23). Brown seaweeds contribute to moisture retention and soil aggregation, due to their high alginate content (24). Red seaweeds, like *Gracilaria*, act as elicitors of plant defence. Green macroalgae, including *Ulva* spp., enhance antioxidant activity and salt stress tolerance as they contain ulvan polysaccharides. This taxonomic diversity is one reason seaweed-based biostimulants remain globally dominant in agriculture (25). Microalgae are microscopic photosynthetic organisms that include genera like *Chlorella*, *Scenedesmus*, *Nannochloropsis*, *Dunaliella salina* and *Spirulina* *Arthrospira*. They contain high levels of protein content (40–70 %), essential amino acids, pigments like chlorophyll a, β -carotene, zeaxanthin and polyunsaturated fatty acids (26). Microalgae play a promising role in precision agriculture and hydroponics by enhancing seed metabolic activation, accelerating germination, stimulating chloroplast development and enhancing water use efficiency (WUE) (27).

Cyanobacteria are increasingly used in multispecies biostimulant consortia along with plant growth promoting rhizobacteria (PGPR), azolla, trichoderma and microalgae. These interactions facilitate improved nutrient cycling, hormone regulation and antioxidant metabolism, thereby improving tolerance of crops to drought and salinity. Such combinations have complementary functions (28). Cyanobacteria and azolla contribute to biological N fixation and organic carbon inputs. The PGPR exhibit 1-aminocyclopropane-1-carboxylate (ACC)-deaminase activity, which is crucial for nutrient solubilisation. It helps in stress mitigation by breaking down the plant stress hormone, ethylene. Trichoderma provides biocontrol benefits and enhances the root strength, while microalgae supply osmoprotectants, pigments and polysaccharides. Together, these organisms strengthen plant resilience and growth during stress (29). Cyanobacteria also play a role in soil restoration by forming microbial crusts rich in exopolysaccharides, which improve soil stability, moisture retention and fertility. This makes them valuable for rehabilitating degraded or saline soils and for promoting climate-resilient agriculture (30). Additionally, the integration of macroalgae, microalgae and cyanobacteria expands application options for biostimulants. Review studies indicate that mixed algal formulations have great potential for sustainable farming and future food security.

Nonetheless, it is important to address the challenges of selecting strains, maintaining formulation stability and validating these methods in large-scale field trials (31). Key bioactive constituents of algal biostimulants and their mechanisms in enhancing plant stress tolerance are shown in Table 1.

Forms and formulations of algal biostimulants

Algal biostimulants are designed to boost metabolite stability, application ease and agronomic performance in a range of cultivation conditions. They are now available in a variety of forms, from conventional seaweed brews to advanced nano-delivery extracts and live microbial formulations. The formulation type influences the field efficacy through its metabolite composition, release pattern and interaction with plant physiology. However, each type of formulation has intrinsic limitations such as variability in metabolite stability, inconsistent field performance in different environments and absence of standardised dose-response relationships, which restricts large-scale reproducibility (32). The biostimulant industry is evolving toward scalable, standardised and precisely controlled products in response to increasing academic interests, offering consistent quality and predictable crop response (33). Forms, formulations and application strategies of algal biostimulants are depicted in Fig. 2.

Liquid extract formulations

Seaweed liquid extracts (SLEs) are prevalent in the commercial algal biostimulant market due to their rapid absorption, convenience for foliar spray applications and compatibility with fertigation systems. Liquid extracts are highly bioavailable, but they have a short shelf life, the bioactive compounds degrade quickly, they are unstable in storage and there is variability between extraction techniques, which can affect consistency of responses in the field (34). Various methods such as hot-water extraction, cold extraction, alkaline hydrolysis, enzymatic hydrolysis, fermentation extraction, supercritical CO₂ extraction, ultrasonication and microwave-assisted extraction are employed to produce these formulations (35). Liquid extracts (0.5–3 %) promote rapid biochemical activation by enhancing photosynthetic pigment synthesis, Rubisco activity, PS II photochemical efficiency, nitrate reductase activity and carbohydrate assimilation (36). They elevate the level of antioxidant enzymes like SOD, CAT, ascorbate peroxidase (APX) and POD, reduce reactive oxygen species (ROS) accumulation, stabilise membranes, maintain osmotic balance and delay senescence under stress conditions (37). These SLEs are widely used in seed biopriming, root dip treatments for nursery seedlings, foliar application, hydroponic nutrient supplements to improve vigour, flowering and nutrient use efficiency (38).

Table 1. Key bioactive constituents of algal biostimulants and their mechanisms in enhancing plant stress tolerance

Bioactive compound	Primary algal source	Physiological/molecular effect	References
Alginate / Fucoidan	Brown macroalgae (ascophyllum, sargassum)	Enhances soil water holding capacity; triggers antioxidant enzyme signalling	(24, 25)
Glycine betaine / proline	Microalgae and macroalgae	Acts as an osmoprotectant; stabilises proteins and membranes	(26, 27)
Phytohormones (auxins, cytokinins)	Green/brown macroalgae, cyanobacteria	Regulates root architecture and delays leaf senescence	(22, 28)
Polysaccharides (laminarin, carrageenan)	Red and brown macroalgae	Elicits defence response (PR-proteins); thickens cell walls via callose deposition	(21, 25)
Phenolic compounds	Microalgae (chlorella), brown macroalgae	ROS scavenging; metal chelation in the rhizosphere	(22, 26)
Exopolysaccharides (EPS)	Cyanobacteria (nostoc, anabaena)	Improves soil aggregation and moisture retention around the root zone	(28, 29)

ROS: reactive oxygen species; PR proteins: pathogenesis-related proteins; EPS: extracellular polysaccharides.

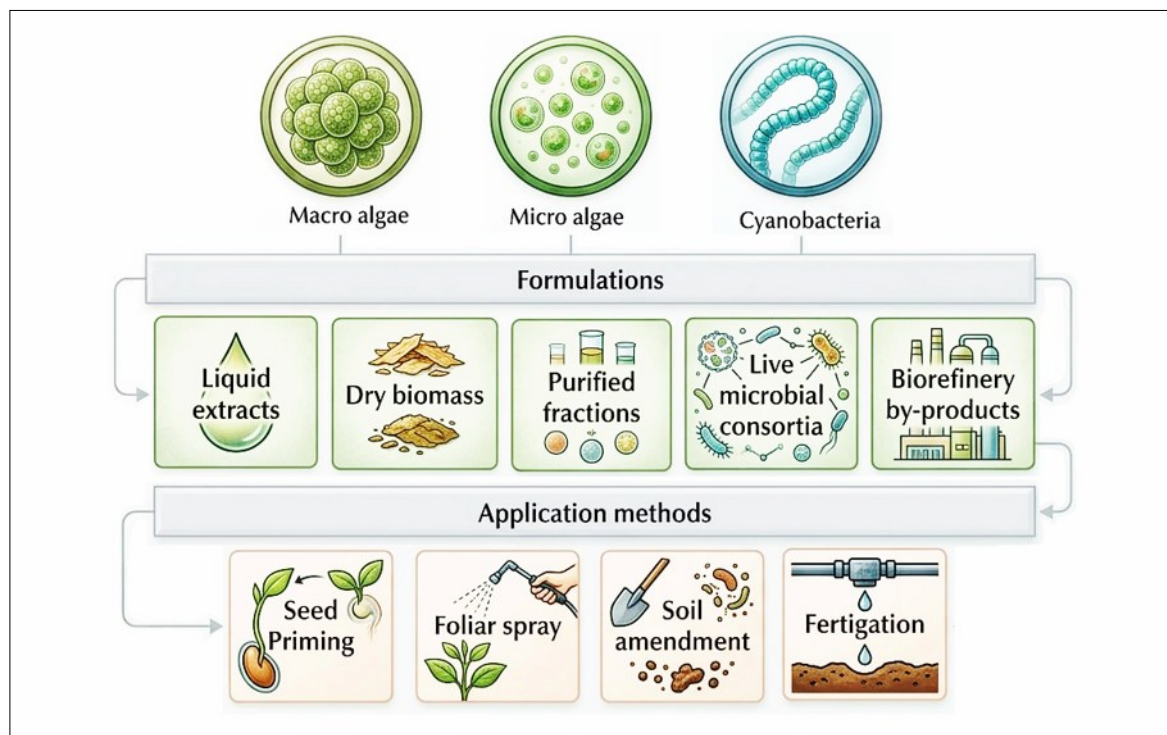


Fig. 2. Forms, formulations and application strategies of algal biostimulants.

Powdered/dry biomass formulations

Powdered/ dry microalgal biomass is produced through sun drying, spray drying, hot-air drying or freeze drying. These formulations enable long shelf life, better transportability and easy bulk application (39). Powdered algae gradually mineralise in soil, supplying macro and micronutrients (potassium (K), calcium (Ca), magnesium (Mg), iron (Fe), zinc (Zn), manganese (Mn)), vitamins, amino acids, fatty acids and polysaccharides, thereby acting as a slow-release carbon-rich amendment (40). Dry biomass formulations of microalgae enhance soil cation exchange capacity, water retention, aggregation and microbial activity in the rhizosphere. Consequently, the product provides prolonged physiological stimulation compared with the quick action of liquid spray formulations. However, drying techniques can result in partial loss of heat sensitive metabolites, decreased enzymatic activity and slow physiological response of plants due to delayed nutrient release (41).

Fractionated and purified bioactive formulations

To achieve specific physiological effects, recent advances focus on isolating high-value biochemicals from algae. Polysaccharides such as laminarin, fucoidan, carrageenan, ulvan and alginate oligosaccharides, enhance tolerance to salinity, pathogens and oxidative stress by activating salicylic acid (SA), jasmonic acid (JA) and ethylene pathways. These molecules strengthen plant immunity through various signalling pathways and lignification of the cell wall. High cost associated with purification and loss of synergistic effects of components, together with limited scalability due to tedious isolation processes, have constrained their widespread use in agriculture, despite their effectiveness (42).

Live microalgal and cyanobacterial formulations

Live algal consortia are highly active biostimulant formulations. These include living microalgae and cyanobacteria that can be introduced directly into the rhizosphere, where they function as nutrient mobilisers, soil conditioners and stress-mitigating agents. They can be applied through seed coating or soil inoculation. These microbes stimulate root growth and improve plant vigour by fixing

atmospheric N and solubilising phosphorus (P). Micronutrient mobility, as well as the release of phytohormones and vitamins, is also enhanced (43). Extracellular polysaccharide biofilms are formed by the cyanobacterial species, which enhance soil aggregation, reduce erosion, improve moisture retention and buffer salinity and alkalinity stress. However, their field performance is limited by low survival under environmental stress, competition with native soil microbes and reduced viability during storage and transport (44).

Biorefinery co-product and circular economy formulations

The rapid expansion of algal based industries, ranging from biofuel production to nutraceuticals, cosmetics and other high value bioproducts, has created a large volume of algal biomass that can be transformed into cost-effective biostimulants (45). It naturally contains high concentrations of carbohydrates, proteins, amino acids, carotenoids, vitamins, polysaccharides and essential mineral salts (46). These compounds enhance soil organic matter, improve microbial activity, increase nutrient solubilisation and stimulate plant metabolic pathways that strengthen growth and stress tolerance. Algal industries promote a circular bioeconomy by converting industrial residues into productive soil amendments. This approach lowers manufacturing costs, environmental degradation and helps in sustainable crop production and more resilient agricultural ecosystems. However, variability in feedstock composition and incomplete removal of industrial impurities can sometimes affect product quality and limit direct agronomic application without further refinement (46, 47). The comparative efficacy of different algal biostimulant forms is given in Table 2.

Key bioactive compounds contributing to stress tolerance

Algal biostimulants have stress-alleviating potential due to their richness source of polysaccharides, phytohormones, betaines, sterols, carotenoids, polyphenolics and minerals, which occur naturally in seaweeds (25). Among these, polysaccharides present in red, green and brown seaweeds, such as laminarin, fucoidan, alginate, carrageans and ulvans, act as potent elicitors that initiate defence signalling in plants, activating pathways linked to pathogenesis-related proteins and oxidative stress management.

Table 2. Comparative efficacy of algal biostimulant forms

Formulation type	Key bioactive components	Mode of application	Advantages	Limitations	References
Liquid extracts	Phytohormones, polysaccharides, antioxidants	Foliar spray, fertigation	Rapid absorption, easy handling	Short shelf life, variability in composition	(46, 47)
Dry biomass	Proteins, minerals, vitamins	Soil amendment, seed coating	Long shelf life, slow release	Lower immediate activity	(36, 45)
Purified fractions	Laminarin, fucoidan, ulvan	Targeted foliar spray	Specific stress tolerance effects	High cost, complex extraction	(33, 35)
Live consortia	Cyanobacteria, microalgae	Soil inoculation, seed coating	Nitrogen fixation, soil restoration	Stability issues, field validation needed	(30, 31, 43, 44)
Biorefinery co-products	Mixed metabolites	Soil amendment	Cost-effective, circular economy	Composition inconsistency	(45, 46)

Laminarin enhances plant defence mechanisms by coding for PR-protein genes (48). Under osmotic stress, alginates enhance soil moisture retention and ionic balance. The phytohormones cytokinin, auxin, gibberellin and abscisic acid (ABA) found in seaweed extracts control stomatal conductance, root growth, chlorophyll retention and stressadaptive morphology, allowing plants to sustain metabolic processes in conditions of drought, salinity and temperature stress (49). Betaines present in seaweeds act as osmoprotectants stabilising membranes and proteins, reducing chlorophyll degradation and improving water retention during salt and drought stress, leading to improved photosynthetic stability and yield (25).

Abiotic stress alleviation by algal biostimulants: Synergistic action of bioactive molecules

The synergistic action of multiple bioactive molecules exerts stress tolerance effects in seaweed treated plants. The synergistic effects of algal biostimulants are not the effect of individual chemicals alone but rather the result of the simultaneous activation of several physiological and molecular pathways. Synergism is achieved by regulating plant responses using an elaborate regulatory process that involves several signalling pathways, including those involving ROS, mitogen activated protein kinase (MAPK) and phytohormones, among others, which enable transcriptional reprogramming and thus facilitate priming under stress conditions (46). Elicitors that induce plant defence signalling pathways include ROS-mediated signalling and MAPK cascades. Polysaccharides such as alginates, laminarins and fucoidans are examples of elicitors. Polysaccharides function as microbe-associated molecular pattern (MAMP) -like elicitors that trigger early signalling for defence response, involving Ca influx, production of ROS and subsequent activation of stress responsive genes (46, 47). Osmo-protectants, such as glycine betaine, are involved in membrane stabilisation and osmotic adjustment, while phytohormones such as auxins and cytokinins regulate root development, nutrient uptake and source-sink relationships. Hormonal modulation promotes interaction among the signalling pathways of ABA, auxins and cytokinins, which improves stomatal regulation and metabolic changes during times of stress (47). These compounds act synergistically to regulate stress perception, signal transduction and physiological adaptation, resulting in enhanced stress tolerance (47).

During extraction, large polysaccharides break into highly bioactive oligomers that act as elicitors, stimulating the signalling pathways that activate the plant's defence network to respond more rapidly to drought, salinity, cold and heat stress (50). Extracts from *Ascophyllum nodosum* have been found to upregulate DREB1A, COR78/RD29A and other cryoprotective genes, increase soluble sugars and enhance unsaturated fatty acids that stabilise membranes under low temperatures and help to tolerate cold/

freezing stress. These effects increase fluidity of the cell membranes and help to stabilise the protein molecules, thus preventing cellular damage from cold shock and retaining enzymatic activity at low temperatures (51).

Similarly, drought-stressed plants treated with seaweed extracts showed higher relative water content, better stomatal regulation, boosted antioxidant enzyme activity and increased expression of ABA-responsive genes, enabling greater physiological resilience under water- deficit conditions. The accumulation of osmolytes such as proline, amino acids and soluble proteins, reduction in oxidative damage and improved recovery after stress exposure are also promoted by these extracts (52).

Under salinity stress, algal extracts activate multiple defence pathways, including upregulation of glutathione S-transferase, late embryogenesis abundant (LEA) proteins, dehydration-responsive genes and ABA signalling components, while simultaneously reducing Na⁺ accumulation and improving K⁺ and Ca²⁺ uptake, thereby restoring ionic balance and maintaining membrane stability (52). The seaweed treated plants show enhanced levels of antioxidant enzymes, including SOD, APX and CAT, lowering ROS levels and preventing oxidative injury to lipids, proteins and DNA. Polyphenolics, carotenoids and lipid mediators like oxylipins support this multi-layered stress defence by strengthening stress signalling networks and redox balance (11).

In general, the interactions between bioactive compounds extracted from algae help in achieving a comprehensive approach to stress adaptation via signal reception, gene expression, osmoprotection, antioxidant protection and metabolic regulation.

Mechanisms by which algal biostimulants enhance stress tolerance

Modulation of plant physiological and biochemical responses

Algal biostimulants stabilise photosynthesis, respiration and overall metabolic activities and play a crucial role in sustaining plant physiological performance under abiotic stresses. Due to stomatal closure during stress conditions such as salinity and drought, chloroplast ultrastructure is disrupted, chlorophyll synthesis is reduced and CO₂ assimilation is limited. Seaweed and microalgal extracts counter these adverse effects by enhancing chlorophyll biosynthesis, protecting photosynthetic pigments from degradation and maintaining higher photochemical efficiency of photosystem II (17). Several studies have shown that the algal biostimulant-treated plants under stress, improved net photosynthetic rate and enhanced stomatal conductance. This physiological stabilisation ensures continuous assimilate supply for growth and stress adaptation, even under prolonged exposure to unfavourable environmental conditions (15,53).

Polysaccharides derived from algae, such as laminarins, carrageenans

and alginates, may facilitate cellular compartmentalisation and reduce electrolyte leakage by stabilising membranes and improving cell wall flexibility.

By minimising irreversible stress damage and promoting healing processes, these coupled biochemical modifications allow plants to maintain metabolic homeostasis (54).

Enhancement of antioxidant systems and reduction of oxidative damage

The overproduction of ROS under abiotic stress conditions results in oxidative harm to lipids, proteins and nucleic acids, surpassing the plant's natural antioxidant capacity. Algal biostimulants can enhance the antioxidant defence mechanisms by stimulating the ROS scavenging processes at both the enzymatic and molecular levels. Multiple studies have demonstrated that, in plants treated with seaweed extracts during salinity and drought stress, the activity of antioxidant enzymes increases, including SOD, CAT, ascorbate peroxidase, glutathione peroxidase and peroxidases. These enzymatic functions effectively transform superoxide radicals into less harmful substances, thereby preventing ROS accumulation and reducing oxidative damage to cellular organelles, especially chloroplasts and mitochondria (55).

The accumulation of non-enzymatic antioxidants is also enhanced by the algal biostimulants, which act as a secondary line of defence against oxidative stress. Algal extracts either directly or through secondary metabolite biosynthesis pathways produce compounds such as ascorbic acid, reduced glutathione, tocopherols, carotenoids, flavonoids and phenolic acids. These molecules scavenge free radicals, protect lipid membranes and stabilise photosynthetic machinery under stress conditions. The enhancement of antioxidant capacity ensures redox homeostasis and prolongs cellular functionality under sustained abiotic stress (56).

Regulation of osmolytes and stress related hormones

Salinity and drought stress lead to osmotic imbalance, which causes cellular dehydration, loss of turgor and enzyme destabilisation. Algal biostimulants stimulate the accumulation of compatible solutes such as proline, glycine betaine, soluble sugars, sugar alcohols and free amino acids, thereby improving the osmotic adjustments. These osmolytes maintain cell turgour, stabilise proteins and membranes and protect macromolecular structures from denaturation. Experimental studies show higher osmolyte concentrations in algal biostimulants treated plants under stress, which correlate with improved water retention, photosynthetic efficiency and growth performance (57).

Algal biostimulants also play a critical role in integrating growth and stress responses by their strong regulatory effects on plant hormonal balance. Seaweed extracts contain hormones such as auxins, cytokinins, gibberellins, ABA analogues and brassinosteroids that regulate endogenous phytohormone biosynthesis and signalling pathways. During stress, algal biostimulants enhance ABA signalling, preventing excessive stomatal closure while maintaining stress sensitivity (48). At the same time, increased auxin and cytokinin activity maintains sink-source connections, delays leaf senescence and promotes root growth. Additionally, interactions with the signalling pathways of SA and JA improve the expression of stress-responsive genes, resulting in increased adaptive plasticity and systemic tolerance (48, 58).

Improvement in nutrient uptake and water use efficiency

Under abiotic stress, nutrient uptake efficiency is severely affected due to reduced root activity, altered membrane permeability and unfavourable soil chemical conditions. Algal biostimulants modify rhizosphere chemistry and stimulate root physiological activity, thereby improving nutrient acquisition. Organic acids, polysaccharides and chelating compounds derived from seaweeds increase nutrient solubility and availability, particularly for P and micronutrients such as Fe, Zn and Mn. Algal treated plants have been reported to show enhanced uptake and translocation of macronutrients including N, P, K, Ca and Mg. Treated plants show enhanced water and nutrient exploration due to their increased root length, lateral root density and root hair proliferation. Algal biostimulants also allow plants to maintain higher transpiration efficiency with reduced water loss and influence stomatal behaviour, aquaporin expression and osmotic regulation. Under water-limited situations, algal-treated plants maintain higher relative water content, better photosynthetic rates and prolonged growth, which eventually leads to improved yield stability and stress tolerance (48). The mechanism of stress tolerance induced by algal biostimulants is depicted in Fig. 3. Comparative summary of agronomic and physiological responses reported in the literature Table. 3.

Algal biostimulants in mitigating specific abiotic stresses exacerbated by climate change

Drought and water scarcity tolerance

Drought stress leads to a decrease in water availability due to unpredictable rainfall and rising temperatures. It significantly impacts plant growth by disrupting cellular hydration and hindering metabolic processes. Algal biostimulants contribute to drought stress tolerance through morphological, physiological and biochemical adjustments. Seaweeds and microalgal extracts notably enhance root system development by increasing root length density, promoting lateral root growth and root hair formation. This improvement enables plants to better acquire water more efficiently under water-scarce conditions. Algal-derived phytohormones like auxin and cytokinin, along with amino acids and polysaccharides, work together to enhance root vigour and functional efficiency (59–61).

In addition to promoting root development, algal biostimulants trigger complex biochemical changes that enhance cellular water retention and resilience against stress. Application of algal extracts leads to increased levels of osmoprotectants such as proline, soluble sugars and glycine betaine, which aid in stabilising proteins and membranes during periods of dehydration and help maintain cell turgour. Supporting this, foliar application of *Ascophyllum nodosum* extract increased soluble protein content from 9.22–11.85 mg g⁻¹ dry weight (DW) and soluble carbohydrate content from 28.18–62.27 mg g⁻¹ DW, indicating enhanced osmotic adjustment under stress conditions (59). Algal treatments mitigate the oxidative stress caused by drought conditions by activating the antioxidant defence mechanisms. For example, *Az nodosum* treatment increased CAT, SOD and APX activities to 12.66, 231.4 and 39.53 $\mu\text{mol min}^{-1}\text{mg}^{-1}\text{protein}$ respectively, while total phenol content reached 168.30 mg catechin acid equivalent (CAE) g⁻¹ DW, demonstrating enhanced antioxidant capacity and stress resilience (52). This comprehensive response enables plants to sustain growth and yield during drought by preserving photosynthetic efficiency and minimising transpiration losses (50).

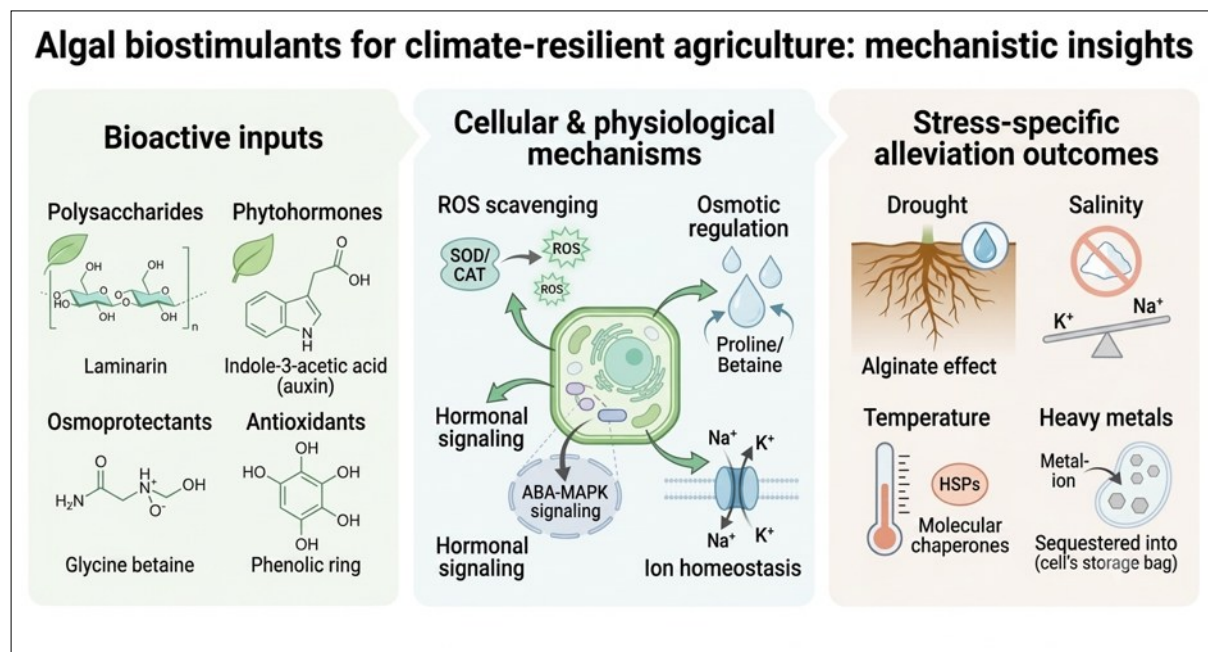


Fig. 3. Algal biostimulants mechanism of stress tolerance.

ROS: reactive oxygen species; ABA: abscisic acid; SOD: superoxide dismutase; CAT: catalase; APX: ascorbate peroxidase; MAPK: mitogen-activated protein kinase.

Table 3. Comparative summary of agronomic and physiological responses reported in the literature

Response domain	Reported outcome trend across studies	Degree of consistency	Evidence pattern across literature	Key influencing factors	References
Biomass production	Positive increase in most studies	Moderate consistency	Strong greenhouse effect, weaker field response	Dose, crop type, formulation	(53)
Photosynthetic efficiency	Increased chlorophyll content & PSII activity	High consistency	Stable improvement under drought/salinity	Stress severity, algal extract type	(15, 36, 53)
Antioxidant system activation	Strong increase in SOD, CAT, APX	Very high consistency	Universal stress response marker	Stress type (all abiotic stresses)	(11, 18, 55)
Yield improvement	Variable (0–40 % increase)	Low–moderate consistency	Strong environment dependency	Field conditions, genotype	(48)
Ion homeostasis (K ⁺ /Na ⁺)	Improved ratio under salinity	High consistency	Strong ionic regulation effect	Salinity level, species sensitivity	(17)
Water use efficiency	Improved but not consistently quantified	Moderate consistency	Mostly drought studies only	Application timing, root traits	(48)
Stress recovery rate	Faster recovery in treated plants	Moderate consistency	Strong early-stage response	Priming vs foliar application	(48, 58)

PSII: photosystem II; SOD: superoxide dismutase; CAT: catalase; APX: ascorbate peroxidase; K⁺: potassium ion; Na⁺: sodium ion.

Salinity stress alleviation

Accumulation of salts in soil at high concentrations adversely affects crop growth and productivity. Salinity induces osmotic stress and ionic toxicity, leading to nutrient imbalance, chlorophyll degradation and overall disruption of metabolic functions. Application of algal biostimulants helps to improve nutrient-use efficiency and regulate ion transport mechanisms, thereby mitigating the adverse effects of saline stress. Algal extracts help in the selective extraction of K and Ca ions, while restricting the sodium accumulation in plant tissues. This maintains favourable K⁺/Na⁺ ratios essential for enzymatic activity and cellular homeostasis. The polysaccharides present in algal extracts improve soil aggregation and water retention and reduce salinity stress in the rhizosphere (62).

Algal biostimulants activate the biochemical pathways at the cellular level and protect the plants from salinity induced oxidative damage. Osmotic balance is maintained under saline conditions by the synthesis of compatible solutes such as proline and soluble sugars. Quantitative studies have shown that regular application of seaweed extracts increased SOD activity by 47–181 %, resulting in enhanced antioxidant protection and improved photosynthetic performance under salt stress (17). The antioxidants derived from

algae reduce the membrane damage caused by the ROS. Thus, it improves the overall chlorophyll stability, photosynthetic activity and biomass accumulation in saline stressed plants (57, 59).

Resilience to temperature extremes (heat and cold)

Under the frequently changing climate conditions, the temperature extremes, particularly heatwaves and cold spells, pose serious threats to crop productivity and survival. High temperature stress reduces photosynthesis and yield by disrupting membrane integrity, protein denaturation and accelerated oxidative damage. Application of algal biostimulants enhances tolerance to high temperature stress by inducing protective molecular responses like upregulation of heat shock proteins (HSPs), which stabilise cellular proteins under high temperature conditions. Furthermore, the bioactive lipids and fatty acids present in the algal extracts help to maintain membrane structure and fluidity and preserve cellular function during elevated temperature stress (57, 61).

Similarly, cold stress affects membrane stability and enzymatic activity and negatively impacts the plant metabolism. The algal biostimulants help in the accumulation of cryoprotective compounds like proline and sugars, which enhance cold tolerance by protecting cells from freeze induced dehydration. Application of

algal biostimulants also regulates the antioxidative defence pathways and reduces the cold stress-induced oxidative damage and chlorophyll degradation. Thus, the physiological and biochemical adjustments made by the algal biostimulants treatment contribute to improved plant survival, faster recovery after temperature stress and enhanced growth in stress conditions (61, 62).

Heavy metal and pollutant stress mitigation

In industrial and intensively cultivated regions, heavy metal contamination and environmental pollution are major concerns for agricultural sustainability and production. Toxic metals such as cadmium (Cd), lead (Pb) and arsenic (As) generate excessive ROS and disrupt the nutrient uptake and inhibit photosynthesis. Polysaccharides, alginates and phenolic compounds present in algal extracts reduce the bioavailability of toxic metals by their strong metal chelating properties. This property reduces the accumulation of metals in edible plant parts and improves food safety. Algal biostimulants stimulate intrinsic detoxification in plants and defence systems. Sequestration of heavy metals into vacuoles is facilitated by the increased synthesis of glutathione and phytochelators, minimising their interference with metabolic processes. Simultaneously, metal induced oxidative stress is neutralised by the antioxidant system (63). Stress-specific mechanisms triggered by algal biostimulants is given in Table 4.

Application strategies for climate-smart use of algal biostimulants

Seed treatment, foliar sprays and soil amendment

Seed treatment using algal biostimulants represents a highly efficient practice for enhancing tolerance to abiotic stress conditions such as erratic rainfall, saline stress and rising temperature extremes. Seed priming with algal extracts controls metabolic activation before germination and enhances the physiological activity of seeds. Various bioactive constituents present in the algal extracts, like phytohormones (auxins, cytokinins, gibberellins), osmoprotectants (betaines), amino acids, vitamins and sulphated polysaccharides regulate enzymatic activity involved in starch and protein hydrolysis, thereby ensuring rapid and uniform germination. Algal priming stabilises cellular membranes, improves ion homeostasis and enhances antioxidant enzyme activity during early seedling growth and help withstand saline and drought stress. In climate stressed conditions, early vigour is crucial, as enhanced root development allows seedlings to access deeper soil moisture and nutrients, improving survival and stand establishment (53).

Foliar application of algal biostimulants provides a rapid and targeted approach to mitigate mid and late- season abiotic stresses. In foliar application, low-molecular-weight algal compounds are directly absorbed through stomata and cuticle (64). In contrast, algal

biostimulants applied as soil amendments provide long term climate adaptation by improving soil physical, chemical and biological properties. Soil aggregation and water holding capacity are enhanced by the algal polysaccharides, while organic acids and bioactive compounds stimulate beneficial microbial populations. Improved soil microbial activity enhances nutrient mineralisation and root-microbe interactions, resulting in sustained crop productivity under climate variability (56).

Timing and dosage optimisation for maximum efficacy

The timing of algal biostimulants application is crucial to its effectiveness as climate smart tool. Early-stage applications, such as seed priming or foliar spraying during early vegetative growth, induce a phenomenon often described as “physiological priming” or “stress memory”. This enables plants to upregulate stress-responsive genes, enhance antioxidant capacity and stabilise hormonal signalling pathways. Thus, it prepares plants to respond more efficiently to stress conditions. As severe abiotic stress causes irreversible cellular damage, pre-stress application of algal biostimulants is more effective than post-stress application. Algal biostimulants can be strategically applied as preventive solutions to enhance crop resilience where stress events such as heat waves and droughts are highly predictable (57).

Dosage optimisation is equally important, as algal biostimulants operate within a narrow physiological response range. Algal extracts stimulate metabolic activity, improve nutrient uptake efficiency and enhance stress tolerance at optimal concentrations. Excessive dosage can cause hormonal imbalance or metabolic inefficiencies. Low to moderate doses produce maximum benefits, while higher concentrations offer diminishing or negative returns. Crop species, growth stage, algal source, extraction method and environmental conditions all influence optimal dosage requirements. Algal biostimulants as climate smart strategies therefore emphasise the need for optimal dosage rather than a single high dose (17).

While microalgae biostimulant applications have demonstrated the ability to alleviate abiotic stresses in plants, results are inconsistent among various studies. Some research reports show that there is a possibility of obtaining similar abiotic stress mitigation responses based on the different physiological processes (31). For instance, some authors state that drought resistance after microalgal application is linked to partial stomatal closure and improved water-use efficiency, whereas other publications report an increase in stomata conductance and photosynthesis. Antioxidant activity has also been shown in many publications; nevertheless, it is not always positively correlated with membrane stability. Another inconsistency pertains to crop qualities, since a higher yield or biomass sometimes coincides with lower quality parameters. This

Table 4. Stress-specific mechanisms triggered by algal biostimulants

Stress type	Key physiological effect	Bioactive molecules involved	Example crop response	References
Drought	Osmotic regulation, stomatal control	Betaines, proline, auxins	Improved water retention in maize	(50, 52, 57, 59–61)
Salinity	Ion homeostasis, ROS reduction	Polysaccharides, antioxidants	Higher K ⁺ /Na ⁺ ratio in rice	(17, 42, 57, 59, 62)
Heat	Protein stabilisation, membrane fluidity	Heat shock proteins, fatty acids	Maintained photosynthesis in wheat	(57, 60, 61)
Cold	Cryoprotective compound accumulation	Sugars, unsaturated fatty acids	Enhanced survival in tomato seedlings	(11, 48, 50, 51)
Heavy Metals	Chelation, detoxification	Phenolics, carotenoids	Reduced oxidative damage in spinach	(11, 55, 63)

ROS: reactive oxygen species; K⁺: potassium ion; Na⁺: sodium ion.

implies that the application of microalgal biostimulants can be highly dependent on the particular microalgal species, formulation, genotype and growing conditions (53).

A significant number of variations among experimental methods of microalgae cultivation and extraction, as well as application procedures, can be another explanation for such inconsistencies between different findings. In addition to that, differences in bioactive compounds due to cultivation and extraction methods can affect the results significantly. Therefore, although the general findings indicate the possible use of microalgal biostimulants in mitigating abiotic stresses, there is a need for more standardisation in characterising the products as well as their field evaluation (31, 53).

Integration with other climate smart practices (e.g., precision agriculture)

The efficiency and scalability of algal biostimulants are enhanced through their integration with other CSA practices. Spatially and temporally precise application of algal formulations is enabled by precision agriculture technologies such as remote sensing, Geographic Information System (GIS)-based soil mapping and decision-support systems. The resource use efficiency is optimised by the application of biostimulants only in stress prone zones detected through vegetation indices or soil moisture sensors. This targeted and timely approach reduces wastage and lowers operational costs. Algal biostimulants can also be used synergistically with other sustainable practices. The combination of algal biostimulants with biofertilisers and microbial inoculants enhance nutrient use efficiency, improves soil carbon sequestration and strengthens overall biological stability. Algal biostimulants reduce the greenhouse gas emission per unit yield, due to reduced dependence on synthetic agrochemicals. Thus, algal biostimulants act as a multifunctional input that supports productivity, resilience and environmental sustainability (65).

Environmental and socioeconomic benefits of algal biostimulants in climate smart agriculture

Reduction in chemical inputs and environmental footprint

Algal biostimulants improve plant nutrient uptake and physiological nutrient utilisation by decreasing reliance on synthetic fertilisers. This allows for lower fertiliser application rates while still maintaining yield (59). The various bioactive compounds found in seaweed and microalgal extracts enhance root development and nutrient absorption from the soil, while also minimising N and P losses due to leaching and volatilisation. The need for frequent fertiliser applications is diminished, leading directly to a decrease in the environmental impact associated with the production, transportation and emissions related to fertilisers (60). The emission of greenhouse gases, especially nitrous oxide, is reduced due to the decreased use of fertiliser inputs.

Enhancement of soil health and biodiversity

Algal biostimulants play an important role in improving the soil aggregation, water holding capacity and soil organic matter content, by the addition of biologically active organic compounds. Soil aggregation, porosity and aeration are improved by the polysaccharides present in algal formulations. Improved soil structure helps the plants to survive under drought prone and climate stressed conditions, by facilitating better infiltration and retention of soil moisture (61).

Soil microbial diversity and activity are enhanced by the algal biostimulants, which serve as substrates and signalling molecules for beneficial microbes like N fixing bacteria, phosphate solubilising microbes and mycorrhizal fungi (6). Soil fertility is improved by diverse microbial species by nutrient cycling, enzymatic activity and organic matter addition (62).

Economic feasibility and scalability for smallholder and commercial farmers

Algal biostimulants serve as an economically feasible approach for improving crop yield when compared to synthetic fertilisers and pesticides. Although the initial costs of biostimulant products may be higher than conventional inputs, they provide higher net returns over growing seasons, due to reduced input frequency and enhanced crop performance. They further reduce the economic risks to farmers by helping maintain stable yield even under climate stress conditions. The quality of the produce is also enhanced, which enables farmers to access premium markets and improve income stability. Due to their low application rates, compatibility with existing farm practices and adaptability across crops, the algal biostimulants can be implemented in both smallholder and large-scale commercial farming systems (66).

Challenges and future directions of algal biostimulants

Standardisation and quality control of algal biostimulants

Lack of standardisation and quality control of the products is the most critical challenge facing the large-scale adoption of the algal biostimulants. The biochemical composition of the algal biomass varies depending on the strain, cultivation conditions, nutrient availability, light intensity, temperature, harvesting stage and extraction method. The bioactive compounds present in algal biostimulants such as phytohormones, polysaccharides, amino acids, vitamins and antioxidants, are influenced by these factors, which results in inconsistent product efficacy under field conditions. Thus, identical application rates of different algal biostimulant products may lead to variable crop responses, limiting commercial scalability (64). In addition, it is also difficult to apply laboratory and greenhouse experiments in the field successfully. Environmental effects along with the variety of crop species and level of stress play an important role in influencing algal biostimulants' effect, making their response quite variable. In addition to that, there are no precise dosage requirements for particular crops and environmental stress factors, which make it difficult to apply them effectively. Extensive field testing is required to develop standardised guidelines for crops (64).

Another major challenge relates to the need to formulate dosage rates for various algae formulations. Algal biostimulants' effectiveness depends largely on the type of formulation, its concentration, mode of application, genotype of the plant, developmental stage and environmental factors. Since the bioactive component in seaweed extract, microalgae formulation, purified fractions and live cells differ, the physiological response observed in the field may vary from one formulation to another (64). Sometimes, using low doses may not generate enough biological effect, whereas too high a dose may cause metabolic disturbances or reduce effectiveness. Consequently, defining dose-response relationship for each specific algae formulation will require conducting field trials at multiple locations (57).

Future efforts must focus on the standardised cultivation protocols, controlled extraction technologies and biochemical fingerprinting of algal products. Advanced analytical techniques such as metabolomics, proteomics and chromatography-based profiling can help establish minimum quality benchmarks. They also help identify key bioactive markers responsible for plant responses. Implementation of good manufacturing practices (GMP), batch-to-batch consistency testing and certification systems will be essential for ensuring the reliability and reproducibility of algal biostimulants. Establishing internationally accepted standards will play a crucial role in improving product credibility and market penetration.

Regulatory frameworks and market acceptance

The absence of harmonised regulatory frameworks at both national and international levels is another major constraint. In many regions, the biostimulants are inconsistently classified as fertilisers, soil conditioners, or plant protection products. This creates ambiguity in registration, labelling and commercialisation procedures. Such uncertainty in regulation increases the compliance costs for manufacturers and slows innovation and market entry (67).

Future strategies emphasise the need for science-based and function-focused regulatory policies that classify algal biostimulants according to their mode of action instead of nutrient content. The European Union biostimulant regulation is one of the recent developments that has a structured framework and could serve as a global model. In addition to these regulatory reforms, enhancing extension services, farmer education programmes and field demonstrations is crucial for effective application and long-term advantages (68).

Research gaps and emerging technologies (Algal biotechnology and nanoformulations)

Considerable knowledge gaps still exist in understanding the molecular and physiological mechanisms of algal biostimulants,

through which they influence plant growth and stress tolerance. Most research primarily examines the short-term growth responses, whereas the long-term impacts on soil health, nutrient cycling, plant-microbe interactions and ecosystem sustainability remain insufficiently explored. Furthermore, the lack of multi-site field trials and the absence of crop and environment specific dosage adjustments hinder the application of laboratory results to real-world agricultural practices.

Future studies are increasingly focused on algal biotechnology and advanced formulation technologies. The production of targeted bioactive compounds can be enhanced through genetic and metabolic engineering, which contribute to specific plant growth promoting functions. Novel nanoformulation strategies, which include nanoencapsulation and controlled-release systems, offer improved stability, increased bioavailability and precise delivery of algal biostimulants under field conditions. Integration of algal biostimulants with precision agriculture tools and climate-smart farming practices helps maximise their efficiency and develop a sustainable agricultural system (69, 70). Additional research also demonstrates that it is essential to integrate molecular mechanisms with large-scale validation to ensure proper functioning of algal biostimulants in changing environments. The use of integrative multi-omics, which includes transcriptomics, proteomics and metabolomics, can help identify important regulatory mechanisms associated with the mitigation of stress factors and can also link the effect of bioactive molecules on plants through molecular mechanisms (71, 72). Identified research gaps and testable hypotheses for future advancement of algal biostimulant research are depicted in Table 5.

Limitations of the present review

The present review is limited by the predominance of studies conducted under controlled conditions, with fewer long-term and multi-location field trials, which limits broader field-level

Table 5. Identified research gaps and testable hypotheses for future advancement of algal biostimulant research

Research gap (critical limitation)	Current limitation in literature	Testable hypothesis (what future studies should prove)	Research priority level	References
Lack of quantitative meta-analysis	No pooled effect size estimation across studies	Algal biostimulants show statistically significant positive pooled effect on yield and stress tolerance across crops	Very high	(53, 65, 66)
Dose standardisation gap	No unified optimal concentration range	There exists a biphasic dose-response curve with optimal stimulation at moderate concentrations	High	(17, 64)
Field validation gap	Over-reliance on controlled conditions	Field-level efficacy is significantly lower than greenhouse results due to environmental variability	Very high	(59, 65)
Mechanistic convergence gap	Multiple pathways reported without hierarchy	ROS-ABA-MAPK signalling cascade is the central conserved mechanism across stress types	High	(18, 58)
Crop-specific response gap	Limited comparative crop studies	Cereal crops rely more on antioxidant pathways, horticultural crops on hormonal regulation	Moderate	(27, 59)
Formulation inconsistency	Lack of standardised bioactive profiling	Fractionated polysaccharides produce stronger and more stable gene expression responses than crude extracts	High	(42, 50)
Soil microbiome interaction gap	Poor understanding of rhizosphere effects	Algal biostimulants enhance stress tolerance through microbiome-mediated nutrient cycling	High	(30, 66)
Long-term sustainability gap	Short-term studies dominate literature	Repeated algal application improves soil carbon and microbial diversity over multi-season cycles	Moderate	(45, 46)

ABA: abscisic acid; MAPK: mitogen-activated protein kinase; ROS: reactive oxygen species.

conclusions. There is also considerable variability in results due to differences in algal sources, extraction methods, formulations, doses, crops and environmental conditions. Furthermore, the absence of standardised reporting and the mechanistic ambiguity under field conditions make direct comparison across studies difficult. This review is based solely on published literature and therefore may also be subject to publication bias and does not include original experimental validation.

Conclusion

Within the scenario of increased variability in climate, algal biostimulants can be considered one of the most promising and sustainable approaches in enhancing the resistance of plants to abiotic stresses in terms of crop productivity. The multiple types of biologically active substances contained in the algal extract have a synergistic effect on plant physiology through regulation of antioxidative defence processes, osmotic adjustments, stomata regulation and enhanced nutrition uptake. In addition to providing additional benefits for crops, the utilisation of algal biostimulants allows for improving soil quality, maintaining balance between the microbial communities within soil and facilitating the implementation of circular bio-economy strategies within agriculture. Nonetheless, in order for this type of biostimulant to become widely used, additional research on its mechanism of action is required, along with development of standard formulations and rigorous field testing at multiple locations.

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Authors' contributions

SS designed the overall review framework and drafted the manuscript. SR supervised the work, refined the focus of the review and provided critical revisions. APS supported literature collection, contributed to concept development, and participated in editing the manuscript. VR helped organise the structure, enhanced the technical quality, and revised the content. MG offered subject expertise and carried out critical evaluation of the manuscript. All authors have read and approved the final manuscript.

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